Ideation Phase

Empathize & Discover

Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Individual project declaration: This project was completed independently by Yuvraj Shrivastava. He independently performed ideation, dataset cleaning and review, Tableau worksheet and dashboard creation, completion of Story Point 1, and website publication. Story Points 2-4 remain planned and were not completed as of 27 October 2026. No other person was involved as a project contributor.

Empathy Map Canvas:

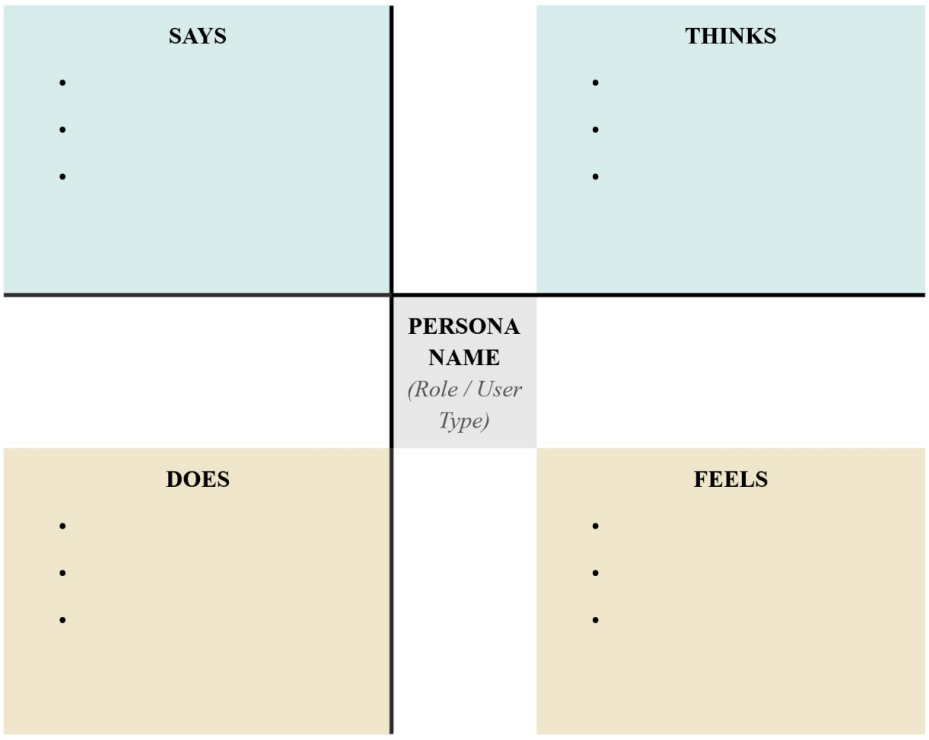
An empathy map is a visual tool that captures a user's behaviours, needs and attitudes. It helps the project creator organize assumptions about the intended user and identify areas that need validation.

It helps the individual project creator consider the user's perspective, goals and challenges before finalizing a solution.

Creating an effective solution requires understanding the problem and the person experiencing it. In this project, Yuvraj Shrivastava prepared a hypothesis-based empathy map for the story persona, Emily Johnson.

Example:

Reference: https://www.mural.co/templates/empathy-map-canvas



Prepared and completed by: Yuvraj Shrivastava | Sole Contributor

COMPLETED EMPATHY MAP

Persona: Emily Johnson | EV Market Strategy Director

Emily Johnson is the persona named in the supplied Tableau Story brief. This map is hypothesis-based, not a record of a verified customer interview.

ROLE / CONTEXT	GOALS
Responsible for evaluating EV market performance and technology opportunities across vehicle categories, manufacturers and countries.	Compare annual sales and manufacturer contribution; understand battery adoption and median range; examine price/range positioning and customer ratings; review manufacturing footprint; communicate strategic recommendations.

SAYS	THINKS
• I need to understand which EV categories contribute most to sales. • How do LFP, NMC, NCA, Sodium-Ion and Solid-State batteries compare? • Which manufacturers and countries should we examine more closely?	• Sales volume is important, but it should be considered with price, range and customer rating. • Battery chemistry and vehicle category may affect product positioning. • A country-level view can help compare the geographic distribution of vehicle records and sales.
DOES	**FEELS**
• Reviews the Executive Summary KPIs and Category Sales chart. • Uses the Manufacturer Share treemap and Global Manufacturing Map. • Compares battery share, median range and the Country vs Battery Matrix. • Examines Price vs Range and Rating vs Battery before communicating the story.	• Wants clear, comparable evidence before proposing a market strategy. • May feel uncertain when a single metric does not explain overall product positioning. • Values an interactive dashboard that makes comparisons easier to explain.

PAINS / CHALLENGES

Market information is distributed across different charts; aggregate KPIs do not explain every category-level difference; battery chemistry comparisons can be difficult to interpret without consistent measures; and every dashboard view must be read in the context of its active filters.

GAINS / EXPECTED VALUE

A single interactive Tableau dashboard; a concise executive summary; side-by-side comparisons of market, technical and geographic measures; and the completed Story Point 1, Global EV Market & OEM Overview. Story Points 2-4 are planned future enhancements.

Data preparation confirmed by the project owner: null values and duplicate records were removed and data types were corrected. ev_market_train.csv was used as the Tableau source; ev_market_test.csv was provided but was not connected to the workbook.

PROJECT EVIDENCE REVIEWED

Screenshots supplied include the dashboard overview, a Story 1 view, the Executive Summary, separate views of worksheets 2–9, and an Excel view of ev_market_train.csv. The worksheet titles visible in the captures are: Executive Summary; Category Sales; Manufacturer Share (Treemap); Battery Performance; Battery Tech Share; Global Manufacturing Map; Price vs Range; Rating vs Battery; and Country vs Battery Matrix.

The Says/Thinks/Does/Feels entries are illustrative persona hypotheses inferred from the project use case. They must not be presented as direct customer quotations or findings from interviews unless actual research evidence is available. The project itself was completed independently by Yuvraj Shrivastava alone; Emily Johnson is only the hypothetical customer persona used for the story.

Appendix A. Visual Context for the Empathy Map

These screenshots provide the Tableau context used to frame the hypothetical EV Market Strategy Director persona. The persona statements remain hypotheses and are not represented as actual interview quotations.

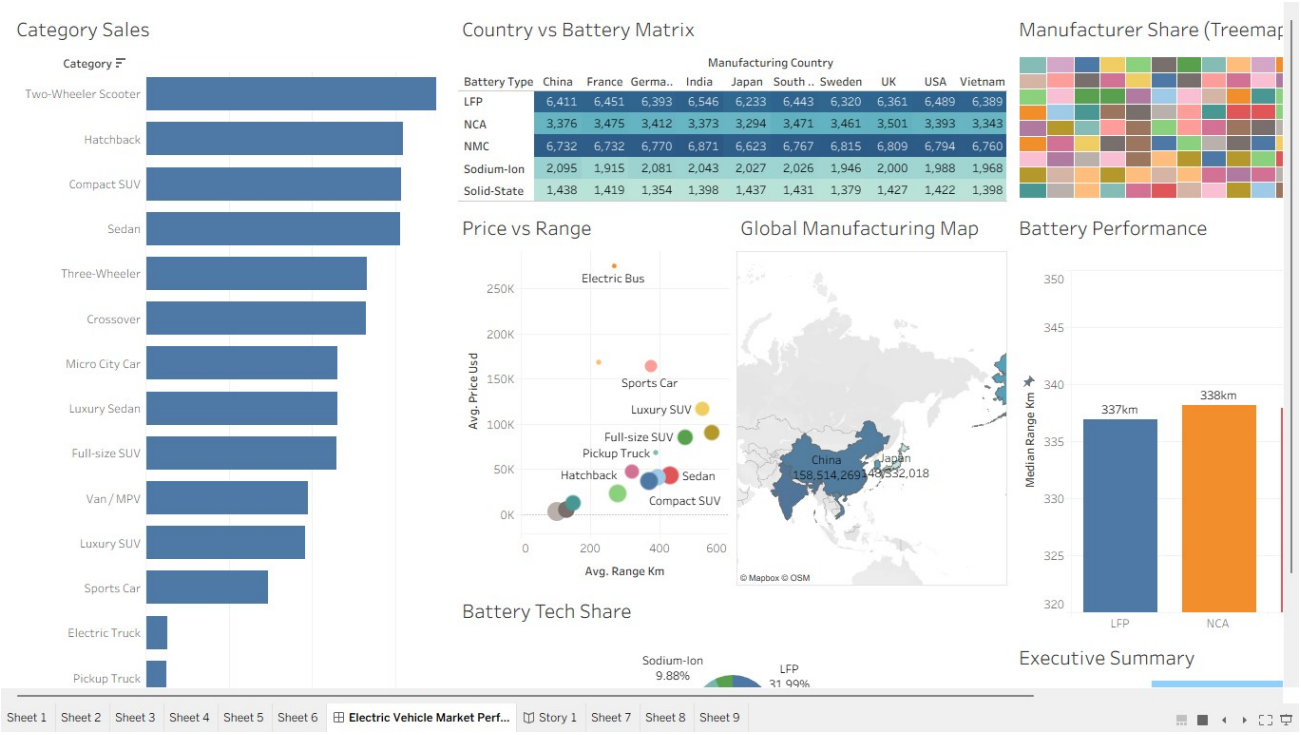


Figure A1. Tableau dashboard overview showing category sales, country-versus-battery matrix, manufacturer treemap, price-versus-range view, global manufacturing map and battery performance.

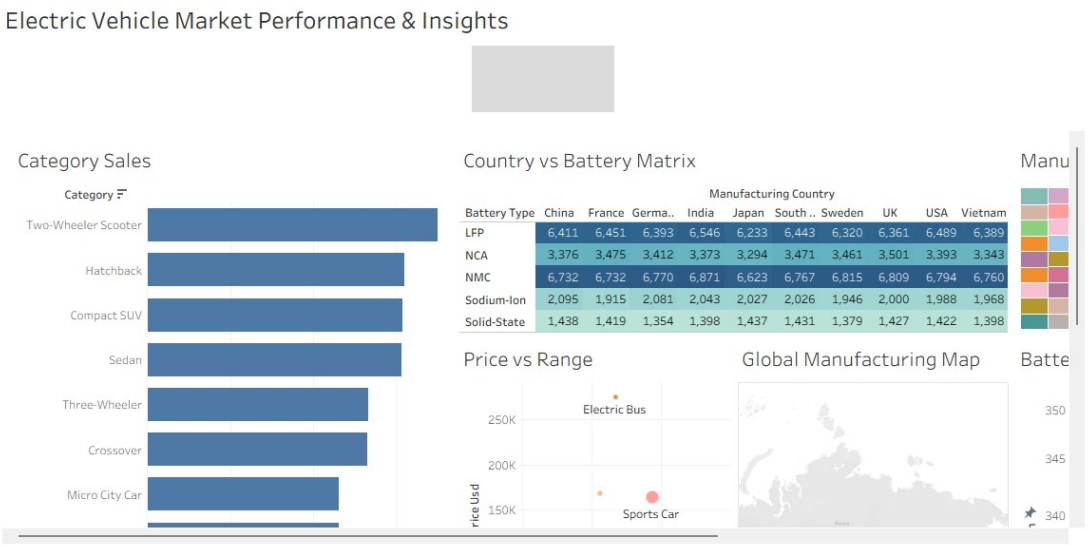


Figure A2. Tableau Story screenshot showing the completed Story Point 1, Global EV Market & OEM Overview. Story Points 2-4 are planned and not yet completed.



Figure A3. Executive Summary worksheet showing the displayed annual sales, average price, average range and average customer rating KPIs.

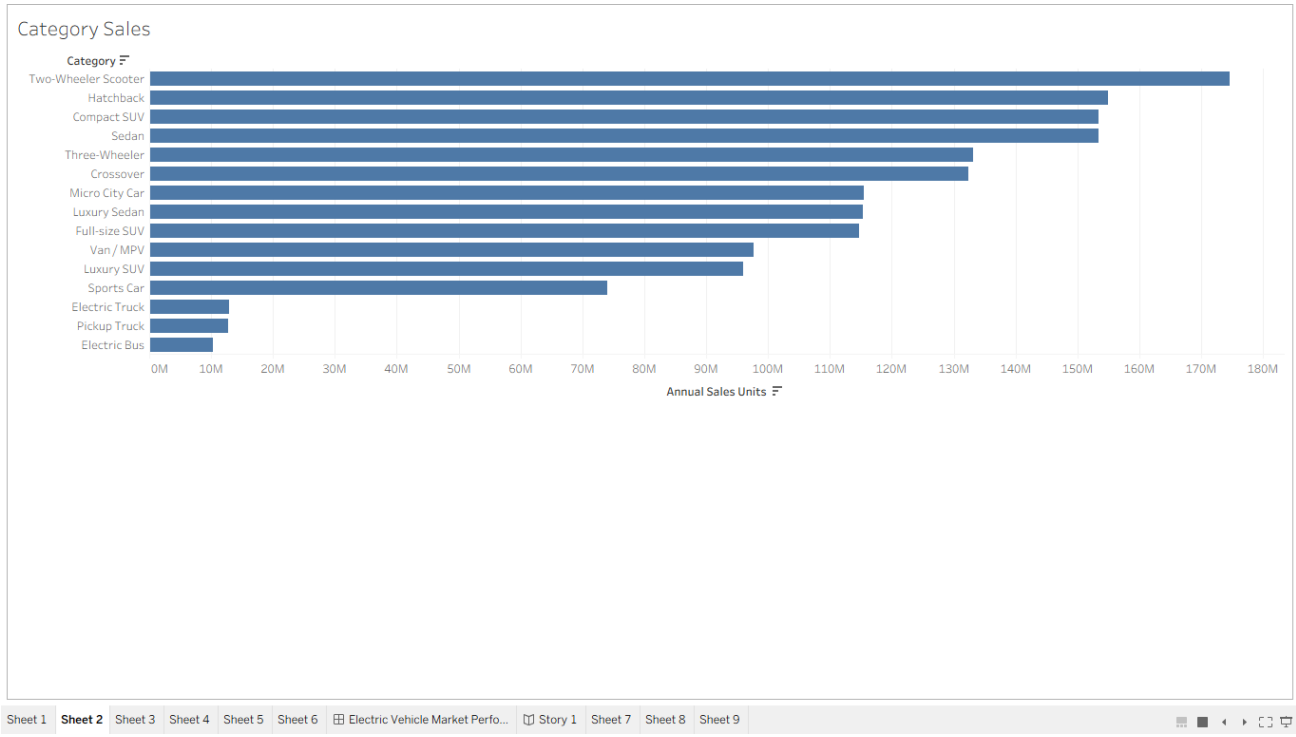


Figure A4. Category Sales worksheet, a horizontal bar chart comparing annual sales units across EV categories.



Figure A5. Battery Tech Share worksheet showing the displayed percentage mix for battery technologies.

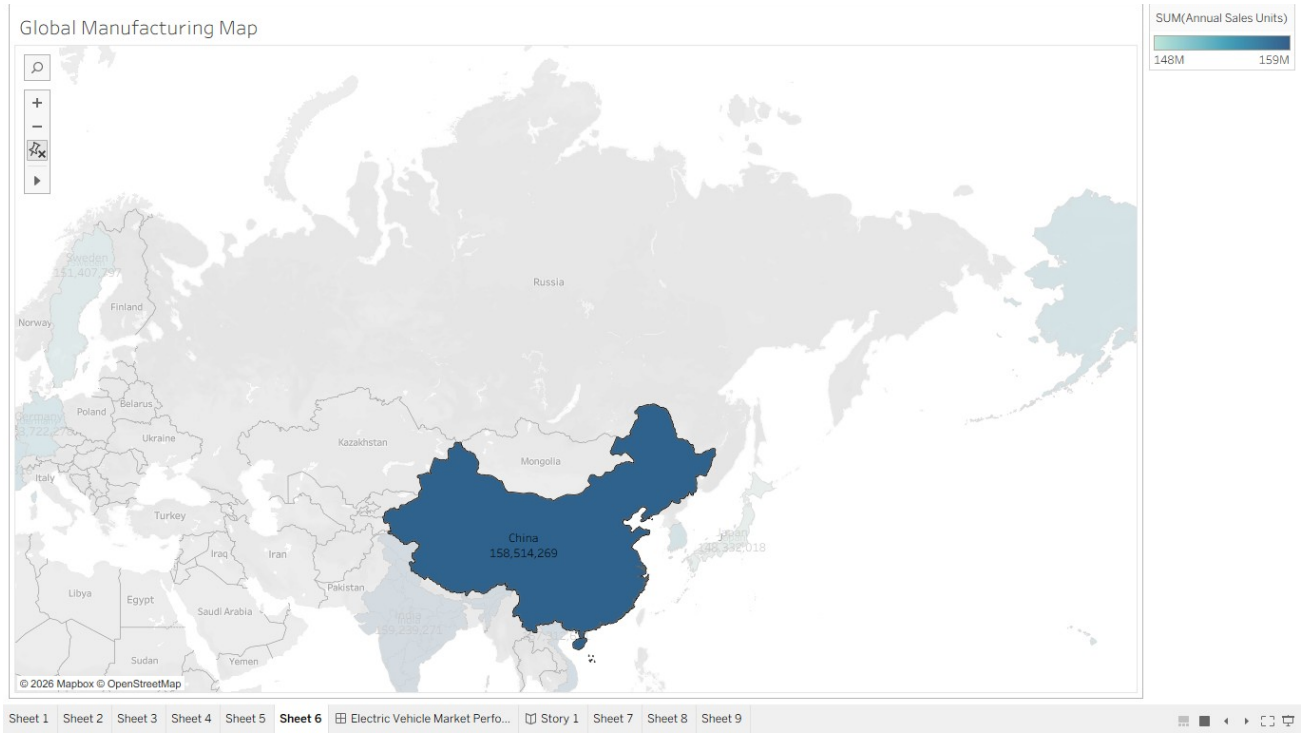


Figure A6. Global Manufacturing Map worksheet, showing geographic sales distribution by manufacturing country.



Figure A7. Price vs Range worksheet, comparing average vehicle range and average price by category.